



Spectral graph fractional Fourier transform for directed graphs and its application

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ABSTRACT

In graph signal processing, the underlying network in many studies is assumed to be undirected. Although the directed graph model is rarely adopted, it is more appropriate for many applications, especially for real-world networks. In this paper, we present a general framework for extending graph signal processing to directed graphs in the graph fractional domain. For this purpose, we consider a new definition for the fractional Hermitian Laplacian matrix on a directed graph and generalize the spectral graph fractional Fourier transform to the directed graph (DGFRFT). Based on our new transform, we then define filtering, which is used to reduce unnecessary noise superimposed on real data. Finally, the denoising performance of the proposed DGFRFT approach is also evaluated through numerical experiments by using real-world directed graphs.

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1. Introduction

In the fields of network science and big data, it is necessary to expand the scope beyond classical time signal analysis and processing to accommodate the signals defined on a graph [1–5]. The irregular structure of the underlying graph is different from the regular structure in classical signal processing, which introduces great challenges to the analysis and processing of graph signals. Graph signal processing (GSP) offers effective tools to process such network data [6–9]. For instance, graph-supported signals can model vehicle trajectories over road networks [10]. Research on GSP has only recently begun, but it is growing rapidly. The main contributions include graph Fourier transform (GFT) and spectral graph wavelet transform (SGWT) [5,11–14], sampling and reconstruction of graph signals [15–19], uncertainty principles [20,21], filtering of graph signals [22,23], etc.

Various transforms of graph signals are still the core of GSP. Among them, GFT acts as a cornerstone [4,5]. In the literature, there are two frameworks for frequency analysis and processing of graph signals: (i) based on the Laplacian matrix [5], and (ii) based on the adjacency matrix [4]. The classical Laplacian-based approach is limited to graph signals located on undirected graphs [5]. In this method, frequency ordering is based on quadratic form and

small eigenvalues correspond to low frequencies and vice versa. Alternately, the adjacency-based approach builds on the shift operator of the graph. This approach constructs the Fourier basis by using generalized eigenvectors of the adjacency matrix. Recently, some unique methods have been proposed to extend the GFT to directed graphs [24–34]. A simpler way is to use the in-degree matrix and weight matrix to define the directed Laplacian [24]. Sardellitti et al. proposed an alternative approach that builds the graph Fourier basis as the set of orthonormal vectors that minimize a continuous extension of the graph cut size, known as the Lovász extension [25]. Sevi et al. introduced a novel harmonic analysis for functions defined on the vertices of a strongly connected directed graph [26]. Shafipour et al. studied the problem of constructing a GFT for directed graphs; they decomposed graph signals into different modes of variation with respect to the underlying network [27]. Then they proposed a methodology [28] to conduct vertex-frequency analyses of graph signals on a directed graph based on [27]. Barrufet et al. also defined a new transform for directed graphs that uses Schur decomposition and leads to a series of embedded invariant subspaces for which orthogonal bases are available [30]. Furutani et al. extended graph signal processing to directed graphs based on the Hermitian Laplacian [31]. Marques et al. provided an overview of the recent landscape of signal processing on directed graphs [32]. Although these methods can construct a Fourier basis with desirable properties, adding some new parameters can balance or enhance the transform coefficients between different vertices, making the analysis more flexible [11,14].

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To explore further developments in graph signals, a new research direction has been proposed in graph signal processing (GSP), that is, fractional order [13,14,23,35–40]. Fractional order has gained considerable attention in the last 20 years in classical signal processing, and the application of fractional order to graphs has also aroused the interest of researchers [23,37–39]. The graph fractional domain is a combination of the graph spectrum domain and the fractional transform domain. The graph fractional Fourier transform (GFRFT) related to graph adjacency matrix was proposed in Wang et al. [13]. Furthermore, a new spectral graph fractional Fourier transform (SGFRFT) related to the graph Laplacian matrix and the spectral graph fractional wavelet transform (SGFRWT) were presented in Wu et al. [14]. By tuning the fractional order, GFRFT and SGFRFT show advantages in revealing the characteristics of the graph signal. However, for directed graphs, both of these transforms have their drawbacks. The Laplacian matrix for SGFRFT is constructed by an undirected graph, so SGFRFT does not apply to directed graphs. Although the GFRFT can be used for directed graphs, it has some potential problems. First, the basis coming from the Jordan decomposition is not orthonormal, Parseval's identity does not hold, and inner products are not preserved in the vertex domain and graph fractional domain. Second, numerical calculations of Jordan decomposition also often produce numerical instability even for medium-sized matrices [41].

The presence of directionality plays a crucial role in modeling social networks, technological networks, biological and biomedical networks, as well as information networks [42–44]. For directed graphs, the existing studies in graph fractional domain are all based on the adjacency matrix [13,35]. In the continuous setting, fractional Fourier transform seeks the orthogonal bases which are eigenfunctions of the fractional Laplacian operator. The background naturally leads us to consider the eigenvectors of the fractional graph Laplacian operator in the discrete setting. Thus, we believe the Laplacian-based construction is more natural. In this paper, we aim to generalize SGFRFT to directed graphs to obtain DGFRFT. The fractional Laplacian operator used in DGFRFT is an extension of the Hermitian Laplacian discussed in Zhang et al. [45] applied to directed graphs, namely Hermitian fractional Laplacian. Moreover, our new DGFRFT links the existing Laplacian-based approach to directed graphs. The paper is organized as follows. We first review the GFT and SGFRFT as our foundation in Section 2. Section 3 introduces the procedure for designing the spectral graph fractional Fourier transform for the signal on a directed graph. Then, an ideal filter and frequency selective filter are presented in Section 4. For the last part, we apply the DGFRFT to signal denoising based on real-world directed graphs and then present experimental results.

2. Preliminaries

2.1. Spectral graph theory

An undirected weighted graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{W}\}$ consists of a finite set of vertices $\mathcal{V} = \{v_0, \dots, v_{N-1}\}$, where $N = |\mathcal{V}|$ is the number of vertices, a set of edges $\mathcal{E} = \{(i, j) | i, j \in \mathcal{V}, j \sim i\} \subseteq \mathcal{V} \times \mathcal{V}$, and a weighted adjacency matrix $\mathcal{W}$. If the values of $\mathcal{W}$ are all in $[0, 1]$ then $\mathcal{W}$ is called an adjacency matrix [44]. $\mathcal{W} = [\mathcal{W}_{ij}] \in \mathbb{R}^{N \times N}$ is defined as $\mathcal{W}_{ij} = w_{ij}$ if $(i, j) \in \mathcal{E}$ and $\mathcal{W}_{ij} = 0$ otherwise. The non-normalized graph Laplacian is a symmetric difference operator $\mathcal{L} = D - \mathcal{W}$ [5], where $D := \text{diag}(d_1, \dots, d_N)$ is a diagonal degree matrix of $\mathcal{G}$, and $d_i := \sum_{j=1}^N w_{ij}$. Let $\{\chi_0, \chi_1, \dots, \chi_{N-1}\}$ be the set of orthonormal eigenvectors. Suppose that the corresponding Laplacian eigenvalues are sorted as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1} := \lambda_{\max}$. Therefore, we have

$$\mathcal{L} = \chi \Lambda \chi^H, \quad (1)$$

where

$$\chi = [\chi_0, \chi_1, \dots, \chi_{N-1}], \quad (2)$$

and the diagonal matrix is $\Lambda = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{N-1}])$. The superscript H represents the conjugate transpose of matrix.

The graph signal f is defined as binding a scalar value to each vertex through the function $f: \mathcal{V} \rightarrow \mathbb{R}$. Using the definition of (inverse) graph Fourier transform (GFT) as in Shuman et al. [5], the GFT of f is

$$\hat{f}(\ell) = \langle f, \chi_\ell \rangle = \sum_{n=0}^{N-1} f(n) \chi_\ell^*(n), \ell = 0, 1, \dots, N-1, \quad (3)$$

where $*$ is complex conjugate. The inverse GFT is given by

$$f(n) = \sum_{\ell=0}^{N-1} \hat{f}(\ell) \chi_\ell(n), n = 0, 1, \dots, N-1. \quad (4)$$

2.2. Spectral graph fractional Fourier transform

The graph fractional Laplacian operator $\mathcal{L}_\alpha$ is defined by $\mathcal{L}_\alpha = \kappa R \kappa^H$, where α is the fractional order, $0 < \alpha \leq 1$ [14]. Note that

$$\kappa = [\kappa_0, \kappa_1, \dots, \kappa_{N-1}] = \chi^\alpha, \quad (5)$$

and

$$R = \text{diag}([r_0, r_1, \dots, r_{N-1}]) = \Lambda^\alpha, \quad (6)$$

so that

$$r_\ell = \lambda_\ell^\alpha, \ell = 0, 1, \dots, N-1. \quad (7)$$

In the follow-up part of this paper, computing the α power of a matrix always uses matrix power function.

The spectral graph fractional Fourier transform (SGFRFT) of any signal f building on graph $\mathcal{G}$ is defined by Wu et al. [14]

$$\hat{f}_\alpha(\ell) = \langle f, \kappa_\ell \rangle = \sum_{n=0}^{N-1} f(n) \kappa_\ell^*(n), \ell = 0, 1, \dots, N-1, \quad (8)$$

when $\alpha = 1$, the SGFRFT degenerates into the standard GFT.

The inverse SGFRFT is given by

$$f(n) = \sum_{\ell=0}^{N-1} \hat{f}_\alpha(\ell) \kappa_\ell(n), n = 0, 1, \dots, N-1. \quad (9)$$

Remark 1. In this paper, to distinguish directed graphs from undirected graphs, we use $\mathcal{G}$ to represent undirected graphs, whereas G represents directed graphs.

3. Spectral graph fractional Fourier transform for directed graph (DGFRFT)

The orthogonality of eigenvectors gives the algebraic ideal properties of the SGFRFT, which makes the graph signal processing of undirected graphs develop well. However, since the eigenvectors of the fractional Laplacian operator of directed graphs are usually not orthonormal, it is difficult to extend graph signal processing to directed graphs simply. Our goal is to find a new way of defining fractional Laplacian matrix to maintain the orthogonality of the eigenvectors and to avoid the calculation of Jordan decomposition, i.e., Hermitian fractional Laplacian matrix. The Hermitian Laplacian is a complex matrix obtained from an extension of the graph Laplacian [31,45–47]. It preserves the edge directionality and Hermitian property. Here we consider a directed graph $G = (V, E, W)$. V is the set of N vertices and E is the set of directed edges. For two vertices $i, j \in V$, $(i, j) \in E$ represents a directed edge from vertex i to j . W is the weighted adjacency matrix of the graph, and its element is defined as $w_{ij} = 1$ if a directed edge (i, j) exists, and $w_{ij} = 0$ otherwise. For directed graphs, the integers d_i^{in} and d_i^{out} specify the number of arrowheads directed toward and away from vertex i respectively [48]. The in-degree of vertex i is calculated as $d_i^{\text{in}} = \sum_{j=1}^N w_{ij}$, whereas, the out-degree is $d_i^{\text{out}} = \sum_{j=1}^N w_{ji}$.

We define a new weighted adjacency matrix as $W_s = [w_{ij,s}]$, where $w_{ij,s} = \frac{1}{2}(w_{ij} + w_{ji})$. $W_s = [w_{ij,s}]$ and E_s (ignoring the directionality of E) uniquely determine the corresponding undirected graph $G_s = (V, E_s, W_s)$ for directed G . The diagonal degree matrix D_s of G_s is $D(i, i) := \sum_{j=1}^N w_{ij,s}$.

Definition 1 (Hermitian graph Laplacian matrix). A Hermitian Laplacian matrix of G is defined as

$$L = D_s - \Gamma_q \odot W_s, \quad (10)$$

where $\odot$ is the Hadamard product [49], and Γ_q is a Hermitian matrix that encodes the edge directionality of G .

Γ_q can take many forms. Here is a simple example. Define $\Gamma_q = [\gamma(i, j)]$ so that it satisfies $\gamma(i, j) = \gamma(i, j)^H$. γ is a map from $V \times V$ to a unitary group of degree 1 and is written as de Resende and Costa [47]

$$\gamma_q(i, j) = e^{2\pi i q(w_{ij} - w_{ji})}, \quad (11)$$

and $0 \leq q < 1$ is a rotation parameter. The parameter q can be selected to make the imaginary part of Γ_q as small as possible, to reduce the amount of calculation and error caused by complex operation. The value of $\gamma_q(i, j)$ represents four possible cases, that is,

$$\gamma_q(i, j) = \begin{cases} e^{2\pi i q} & \text{if } (i, j) \in E, \text{ i.e. } w_{ij} - w_{ji} = 1; \\ e^{-2\pi i q} & \text{if } (j, i) \in E, \text{ i.e. } w_{ji} - w_{ij} = 1; \\ 1 & \text{if } (i, j) \text{ and } (j, i) \in E, \text{ i.e. } w_{ij} = w_{ji} = 1; \\ 1 & \text{no edge between } i \text{ and } j, \text{ i.e. } w_{ij} = w_{ji} = 0. \end{cases} \quad (12)$$

$\gamma_q(i, j) = e^{2\pi i q}$ means the edge from vertex i to j . This definition also covers mixed graphs. If $(i, j), (j, i) \in E$, that is, $w_{ij} - w_{ji} = 0$, it can be viewed as a single undirected edge. When $\gamma_q(i, j) = 1$, the existence of edges is further distinguished after taking the Hadamard product with W_s .

As D_s and W_s are real symmetric matrices, L is a Hermitian matrix. Let v_k and u_k be the k th eigenvalue and eigenvector of the Hermitian Laplacian L , respectively. The eigendecomposition of L can be written as

$$L = UVU^H, \quad (13)$$

where $U = [u_0, u_1, \dots, u_{N-1}]$. $V = \text{diag}([v_0, v_1, \dots, v_{N-1}])$ is a real diagonal matrix, and Hermitian Laplacian eigenvalues are sorted as $0 \leq v_0 \leq v_1 \leq v_2 \leq \dots \leq v_{N-1} := v_{\max}$.

The graph Hermitian fractional Laplacian matrix for a directed graph is given by $L_{\alpha,d} = PQP^H$, in which

$$P = [p_0, p_1, \dots, p_{N-1}] = U^\alpha, \quad (14)$$

and

$$Q = \text{diag}([\xi_0, \xi_1, \dots, \xi_{N-1}]) = V^\alpha, \quad (15)$$

that is

$$\xi_\ell = v_\ell^\alpha, \quad \ell = 0, 1, \dots, N-1. \quad (16)$$

The total variation for graph signals is defined as an absolute sum of the discrete difference of a graph signal [50]. It was first introduced as an extension of the original total variation (TV) [51]. We define the total variation of a graph signal f as

$$\text{TV}(f) := \sum_{(i,j) \in E} |f(i) - f(j)|^2 \quad (17)$$

to measure the smoothness of signals. The total variation has an intuitive interpretation: it compares how the signal varies with time or space and calculates the cumulative magnitude of the signal. The smaller the difference between the original signal $f(i)$ and $f(j)$, the lower the signal's variation.

To describe some spectral characteristics of Hermitian fractional Laplacian matrix $L_{\alpha,d}$, we calculate eigenvalues and the total variation of the Hermitian fractional Laplacian matrix for five different values of q and two different α ($\alpha = 0.8$ and 0.6). The underlying graph we use is a random directed graph constructed by fixing k vertices and generating a direct edge with probability p for each pair of vertices ($k = 50, p = .1$). The eigenvalues are presented in Fig. 1. Fig. 2 shows the total variations of eigenvectors of $L_{\alpha,d}$. When the rotation parameter $q = 0$, the Hermitian fractional Laplacian matrix degrades to a fractional Laplacian matrix as each element of Γ_q equals 1. Therefore, in Fig. 1, when the values of q are relatively small, the spectrum of $L_{\alpha,d}$ is similar to the spectrum of fractional Laplacian matrix. The orange line and the blue line illustrate this feature. As q increases, the spectrum is different from $q = 0$ as the green line shows in the zoomed-in version in Fig. 1. Fig. 2 also shares a similar conclusion, and there is oscillation when q is large. Furthermore, the oscillations become more pronounced as the value of α decreases.

Then we prove that $L_{\alpha,d}$ is a positive semi-definite Hermitian matrix.

Proposition 1. For any fractional order α , $L_{\alpha,d}$ is a Hermitian matrix that satisfies

$$(L_{\alpha,d})^H = L_{\alpha,d}. \quad (18)$$

Proof. From the definition of $L_{\alpha,d}$, we have

$$(L_{\alpha,d})^H = (PQP^H)^H = (U^\alpha V^\alpha (U^\alpha)^H)^H = U^\alpha (V^\alpha)^H (U^\alpha)^H. \quad (19)$$

Since V is a real diagonal matrix, i.e., $(V^\alpha)^H = V^\alpha$. Therefore, we obtain

$$(L_{\alpha,d})^H = U^\alpha (V^\alpha)^H (U^\alpha)^H = U^\alpha V^\alpha (U^\alpha)^H = PQP^H = L_{\alpha,d}. \quad (20)$$

□

Proposition 2. For any fractional order α , $L_{\alpha,d}$ is a positive semi-definite matrix.

Proof. From Proposition 1, we know that V^α is a real diagonal matrix. Let $J = V^{\alpha/2} (U^\alpha)^H$, so $L_{\alpha,d} = J^H J$. For any signal $f \in \mathbb{C}^N$, we note that

$$f^H L_{\alpha,d} f = f^H U^\alpha V^\alpha (U^\alpha)^H f = f^H J^H J f = (Jf)^H Jf \geq 0. \quad (21)$$

□

Clearly, in this case, this Hermitian fractional Laplacian matrix has a set of orthonormal eigenvectors. This orthogonality allows the basic concepts of graph signal processing for undirected graphs to be extended directly to those of directed graphs.

Definition 2. (Directed graph fractional Fourier transform) The spectral graph Fractional Fourier Transform for directed graph (DGRFT) of any signal f building on graph G is defined as

$$\hat{f}_{\alpha,d}(\ell) = \langle f, p_\ell \rangle = \sum_{n=1}^N f(n) p_\ell^*(n), \quad \ell = 0, 1, \dots, N-1. \quad (22)$$

By the matrix form, the DGRFT is

$$\hat{f}_{\alpha,d} = P^H f. \quad (23)$$

The inverse DGRFT is given by

$$f(n) = \sum_{\ell=0}^{N-1} \hat{f}_{\alpha,d}(\ell) p_\ell(n), \quad n = 0, 1, \dots, N-1. \quad (24)$$

Now we have the definition of DGRFT, which has some useful properties.

Property 1. We can check that the DGRFT matrix is unitary, that is,

$$P^{-1} = (U^\alpha)^{-1} = U^{-\alpha} = P^H. \quad (25)$$

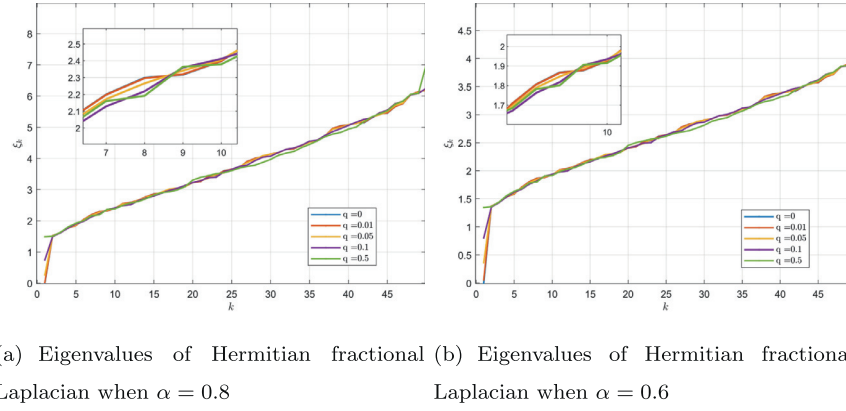


Fig. 1. Eigenvalues of Hermitian fractional Laplacian with respect to different fractional orders.

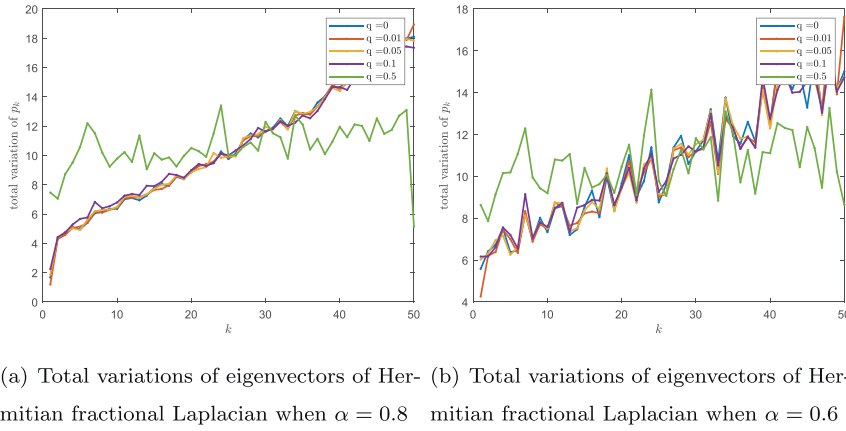


Fig. 2. Total variations of eigenvectors of Hermitian fractional Laplacian with respect to different fractional orders.

Property 2. It is easy to check that the DGRFT matrix satisfies index additivity. Given two different fractional orders α and β , we have

$$U^{-\alpha} \circ U^{-\beta} = U^{-\beta} \circ U^{-\alpha} = U^{-(\alpha+\beta)}.$$

Property 3. When $w_{ij} = w_{ji}$, our DGRFT reverts to SGFRFT defined in Wu et al. [14], which corresponds to treating the graph as undirected. Then, for $\alpha = 1$, the SGFRFT is no more than the GFT defined by Shuman et al. [5].

Property 4. The Parseval relation holds. For any signal f and g defined on the directed graph G , we have

$$\langle f, g \rangle = \langle \hat{f}_{\alpha,d}, \hat{g}_{\alpha,d} \rangle. \quad (26)$$

If $f = g$, then

$$\begin{aligned} \sum_{n=1}^N |f(n)|^2 &= \|f\|_2^2 = \langle f, f \rangle \\ &= \langle \hat{f}_{\alpha,d}, \hat{f}_{\alpha,d} \rangle = \|\hat{f}_{\alpha,d}\|_2^2 = \sum_{\ell=0}^{N-1} |\hat{f}_{\alpha,d}(\ell)|^2. \end{aligned} \quad (27)$$

DGRFT is a new transform focused on directed graphs in graph fractional domain. Compared with GFRFT, it has several significant advantages. First, the columns of U are linearly independent eigenvectors, and at the same time they are orthogonal. This results in DGRFT that preserves the inner product when passed from the vertex domain to the graph fractional domain. In addition, DGRFT preserves edge directionality. Finally, computing the GFRFT needs to do Jordan decomposition. When the size of the graph exceeds the median value, the calculation of Jordan decomposition leads to serious and difficult numerical instability. As our new Laplacian matrix for directed graphs is Hermitian matrix, the Jordan decomposition can be avoided in the calculation of DGRFT.

Remark 2. It should be noted that the definition of DGRFT and its inverse depends on the choice of eigenvectors of Hermitian fractional Laplacian matrix, which is not unique. How to choose the ideal eigenvector to optimize the analysis is still an open question. The readers can choose the appropriate eigendecomposition of the Hermitian fractional Laplacian matrix based on their experience. The actual application background, the computational cost, the ease of implementation, and other potential factors can be taken into account.

Table 1 gives a brief summary of the fundamental transforms based on spectral graph theory in GSP. The column of the basic building block gives the origin of the definitions for five different transforms, and the last column shows the graph structure suitable for the above transforms.

4. Directed graph filtering

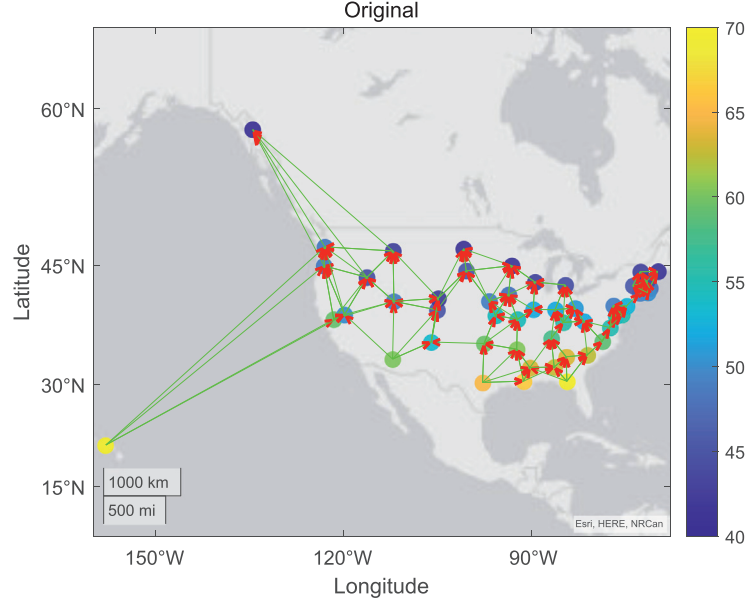
4.1. Spectral graph filtering

In classical signal processing, filtering can be defined by convolution. The convolution of signal a with b is the result of signal a filtered by b . Therefore, to define directed graph filtering in graph fractional domain, we first need to define the convolution operator. Convolution in the time domain is equivalent to multiplication in the Fourier domain. For a directed graph, the graph fractional convolution operator is defined as the following form consistent with the classical convolution by using the Hermitian fractional Laplacian eigenvectors.

Table 1

Some fundamental transforms based on spectral graph theory and their underlying graph structure in GSP.

	GFT [5]	SGFRFT [14]	DGFRFT	SGWT [11]	SGFRWT [14]
Basic building block	graph Laplacian matrix	graph fractional Laplacian matrix	graph Hermitian fractional Laplacian matrix	a kernel and graph Laplacian matrix	a fractional kernel and graph fractional Laplacian matrix
Underlying graph structure	undirected	undirected	directed	undirected	undirected

**Fig. 3.** Average temperature data measured by 50 meteorological state stations in the US.

Definition 3 (Convolution operator). For any graph signal f and g , whose underlying graph structure is directed, their graph fractional convolution $*$ is

$$(f * g)(n) = \sum_{l=0}^{N-1} \hat{f}_{\alpha,d}(l) \hat{g}_{\alpha,d}(l) p_l(n). \quad (28)$$

A graph filter is a system that takes a graph signal as input and produces another graph signal as output [52]. Given an input directed graph signal f_{in} , the filtering is defined by the convolution of f_{in} and a filter h . Thus the spectral directed graph filtering in vertex domain is

$$f^{out}(n) = (f * h)(n) = \sum_{\ell=0}^{N-1} \hat{f}_{\alpha,d}^{in}(\ell) \hat{h}_{\alpha,d}(\ell) p_\ell(n), \quad n = 0, 1, \dots, N-1, \quad (29)$$

and in graph fractional domain

$$\hat{f}_{\alpha,d}^{out}(\ell) = \hat{f}_{\alpha,d}^{in}(\ell) \hat{h}_{\alpha,d}(\ell), \quad \ell = 0, 1, \dots, N-1. \quad (30)$$

4.2. Frequency selective filtering

The above spectral graph filtering is an ideal filter. It retains all frequencies within a given range and completely removes those that are out of range. Because the graph has a finite number of points, it is impossible to design such a filter. A frequency selective filter is a system that isolates the specific frequency components and excludes all others. It is easier to achieve in application. The ideal response of the frequency selective filter is [53]

$$\hat{h}_d(l) = \begin{cases} 1, & l \in \text{passband} \\ 0, & l \in \text{stopband}. \end{cases} \quad (31)$$

Lowpass, bandpass and highpass filters are three common frequency selective filters.

Using $\hat{h}_d(l)$, the graph signal f is filtered in graph fractional domain by

$$\hat{f}_{\alpha,d}^{out} = J_d \hat{f}_{\alpha,d}, \quad (32)$$

where $J_d = \text{diag}([\hat{h}_d(\lambda_0), \hat{h}_d(\lambda_1), \dots, \hat{h}_d(\lambda_{N-1})])$.

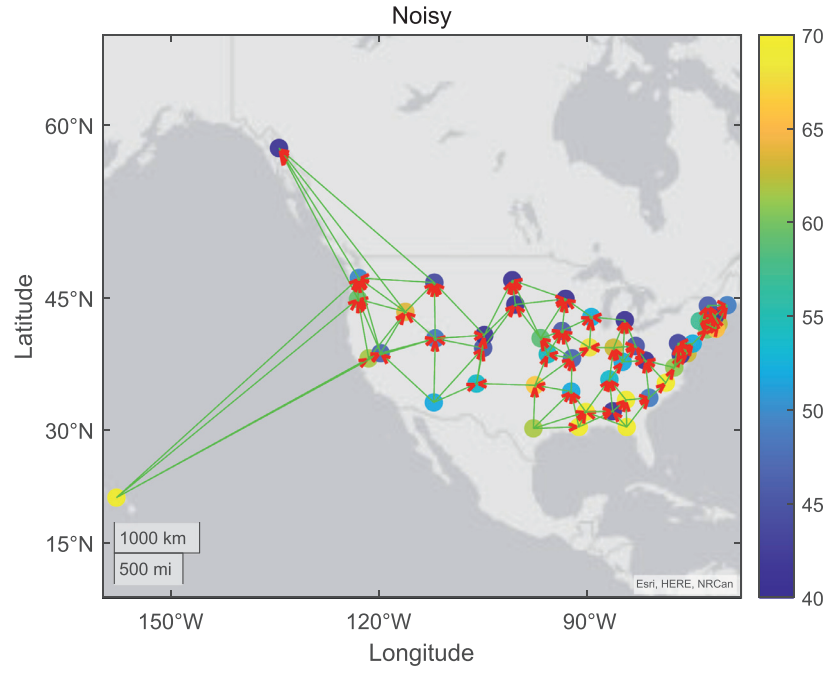
Or, equivalently, vertex domain filtering can be obtained by the inverse DGFRFT as follows

$$f^{out} = P \hat{f}_{\alpha,d}^{out} = P J_d \hat{f}_{\alpha,d} = P J_d P^H f = H f, \quad (33)$$

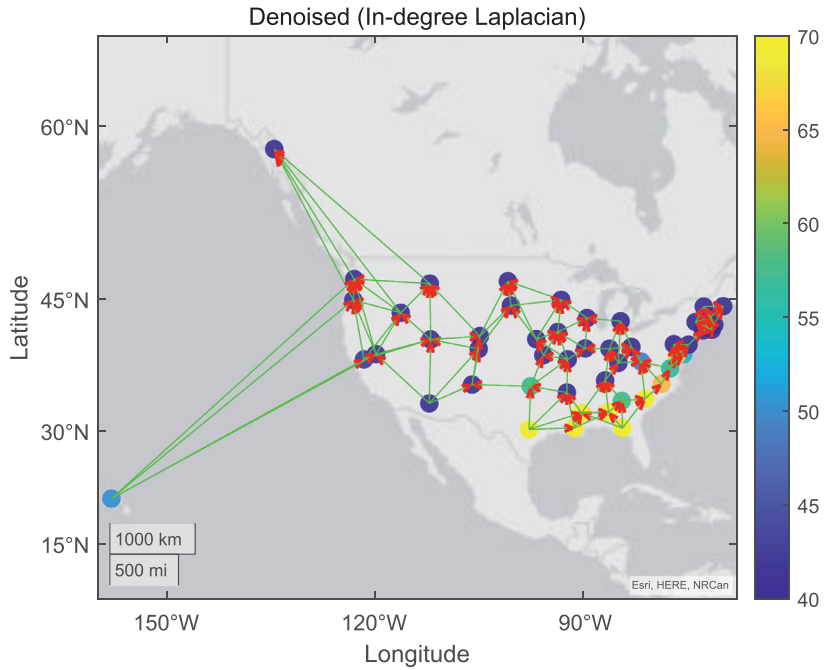
where $H = P J_d P^H$ represents the transfer matrix.

5. Experimental results

Here we assess the performance of DGFRFT via simulations on three graphs. In Simulation 1, we first use the US temperature data to verify that directionality is an important factor in application. For a graph that starts out as a directed graph, we apply DGFRFT for denoising. Then, the directionality of the graph is ignored, the same denoising step is performed, and the results are observed. Two transforms of directed graph and another two transforms of undirected graph are selected for comparison with DGFRFT. In addition, evaluating the denoising performance of our DGFRFT on larger graphs is necessary. We give an processing example on Wikipedia voting data that contains 7115 vertices and 103,689 directed edges. In graph fractional domain, DGFRFT and GFRFT [13] are suitable for directed graphs. The former is based on Laplacian matrix, and the latter uses adjacency matrix. In Simulation 2, for a real dataset, we use DGFRFT and GFRFT to perform the same filtering and denoising task, and then compare the results. The experimental results show that our DGFRFT introduces



(a) noisy signal.



(b) denoised signal using in-degree GFT defined in [24].

Fig. 4. Noisy and denoised temperature signals on directed US graph.

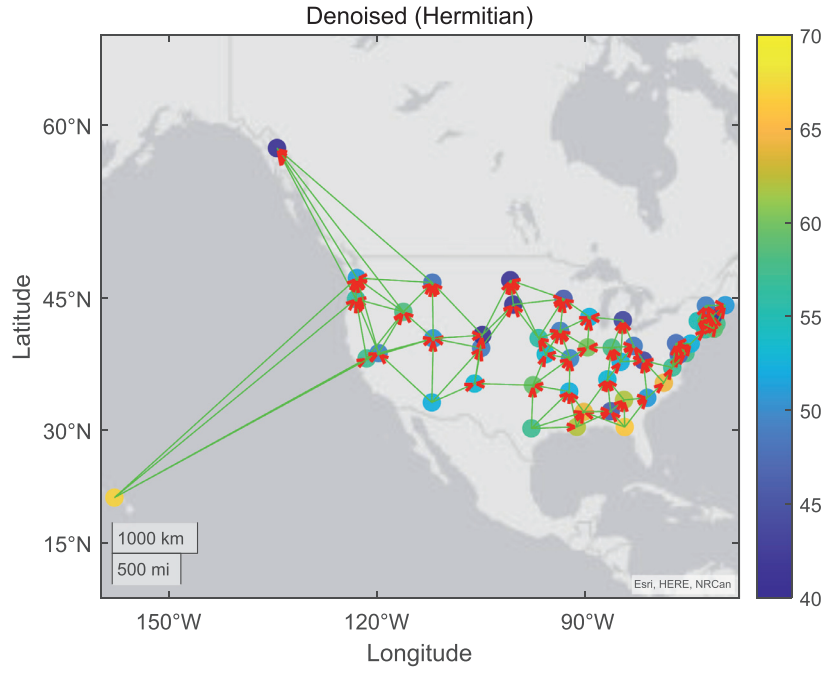
smaller errors and has higher robustness. All simulations are performed on a laptop with Intel Core i7-10710U, 16 G RAM, and MATLAB R2021b.

5.1. Simulation 1: directed vs. undirected

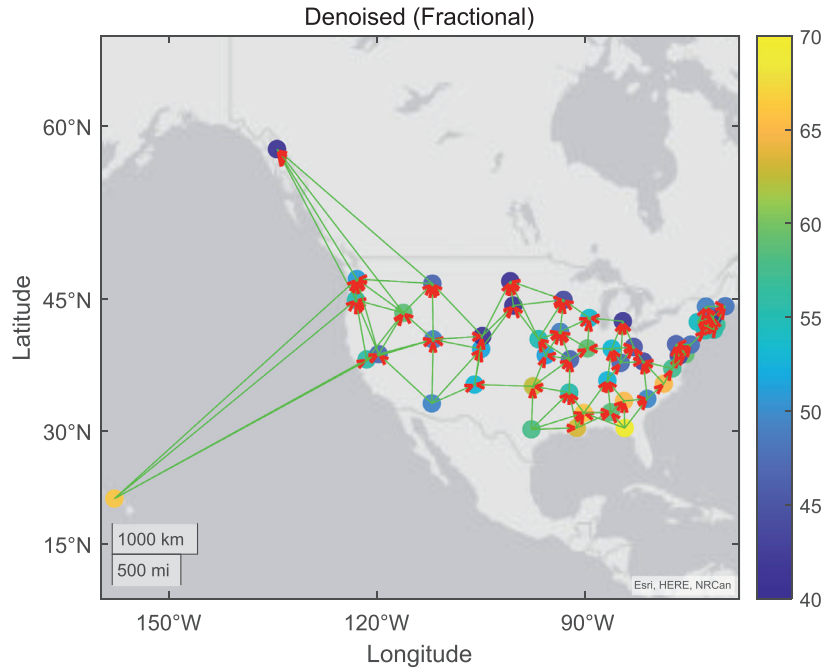
5.1.1. On small-scale graphs

In this section, real temperature data measured by 50 state meteorological stations in the United States (US) are used to conduct experiments. We consider signal denoising on a directed graph. First, based on a map of the US, we define a directed graph by

using geographic locations and latitude. In this graph, the vertices represent the 50 states of the US. The average annual temperature in each state is viewed as a graph signal. Only when two states share a border are they connected by an edge. The direction of an edge is based on latitudinal assignments from states with low latitude to states with high latitude. Latitude is an important factor affecting the temperature. Temperature is inversely related to latitude. As latitude increases, the temperature falls, and vice versa. Thus, there is a correlation between latitude and temperature and it is justified to assign directed edges according to state latitude to capture the temperature flow. The



(c) denoised signal using Hermitian GFT [31].



(d) denoised signal using DGFRFT.

Fig. 4. Continued

same setting is also used in some literature, such as [31,32,54]. Fig. 3 shows the US temperature signal on a directed graph. The data is available at <https://www.currentresults.com/Weather/US/average-annual-state-temperatures.php>.

We generate a noisy signal $g = f + n$, where f is the original signal representing the average temperature on the graph and n is the noise vector whose coordinates are independently sampled from the Gaussian distribution with zero mean and standard deviation $\sigma = 10$. We apply classical GFT [4], SGFRFT [14], another kind of GFT defined by in-degree matrix [24], Hermitian GFT [31], and

DGFRFT to recover the original signal from the noisy signal. For each method, we use a low-pass filter kernel

$$\hat{h}_d(\lambda) = \frac{1}{1 + c\lambda}, \quad (34)$$

where λ is the eigenvalue corresponding to classical graph Laplacian, graph fractional Laplacian, in-degree graph Laplacian, Hermitian graph Laplacian, and graph Hermitian fractional Laplacian. The reason we choose this filter is that we want to find an easy yet meaningful low-pass filter to use. This low-pass filter

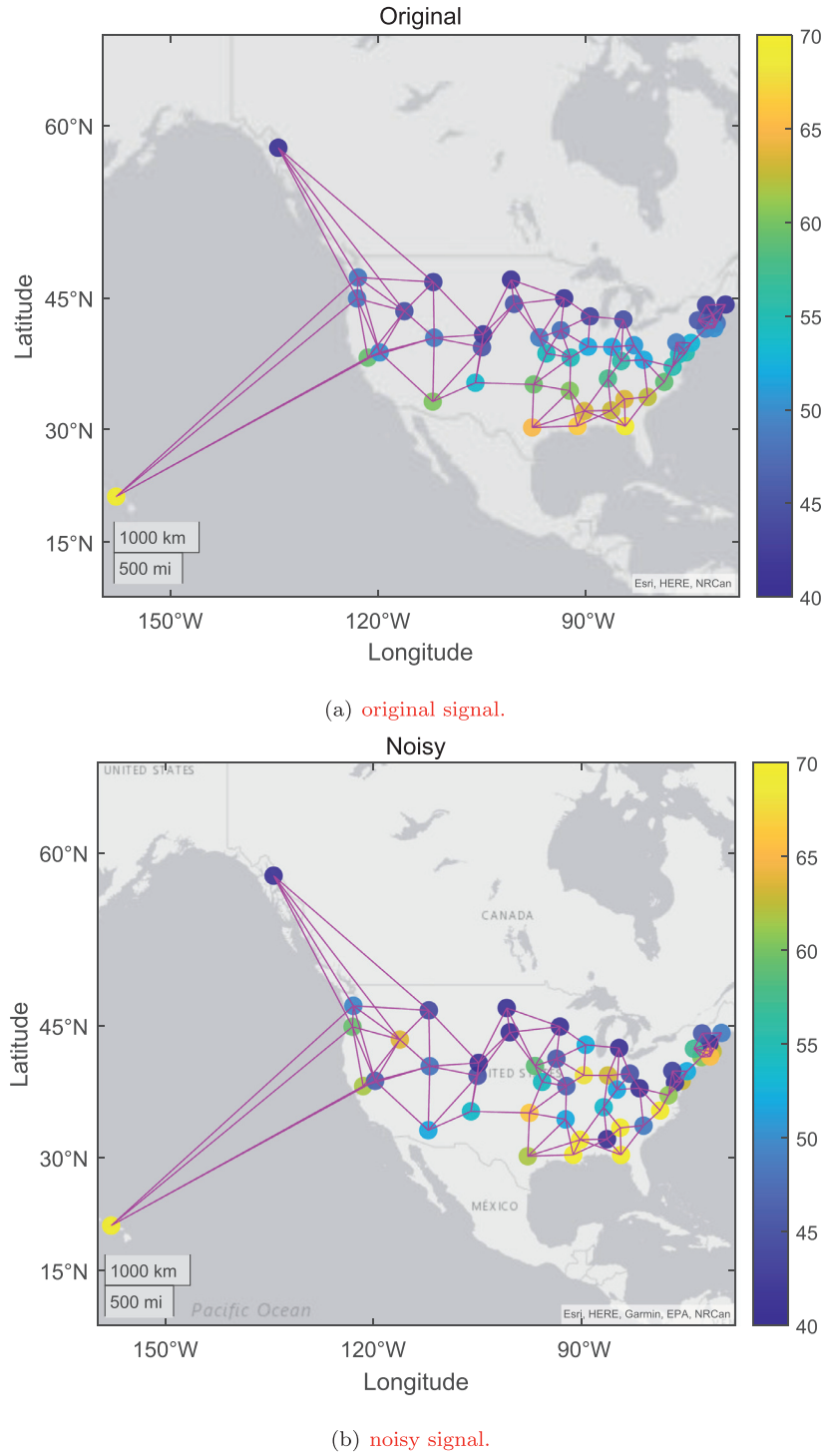


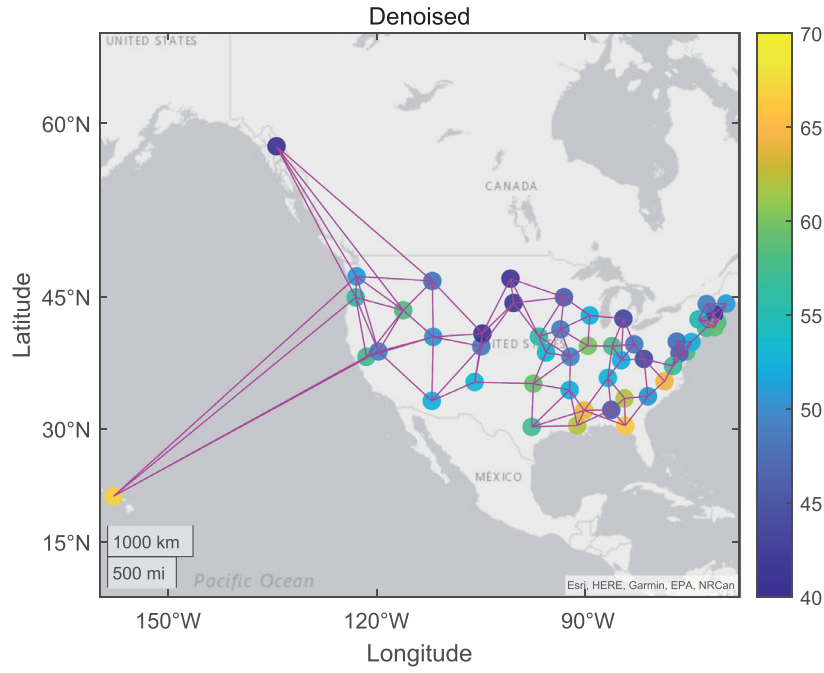
Fig. 5. Original, noisy and denoised temperature signals on undirected US graph.

should have an explicit form, and contain as few tuning parameters as possible. The simplest form is a linear function $c\lambda + b$. To suppress high frequencies we invert the linear function $1/(c\lambda + b)$ so that when λ is large the coefficient of the filter tends to 0. To let low frequencies pass we let $b = 1$ so that when λ is small the coefficient of the filter tends to 1. The constant c is the only tuning parameter and controls the strength of the filter. If c is large then only very low frequencies can pass. Conversely, if c is small then the filter is tolerant with high frequencies. The denoised signal

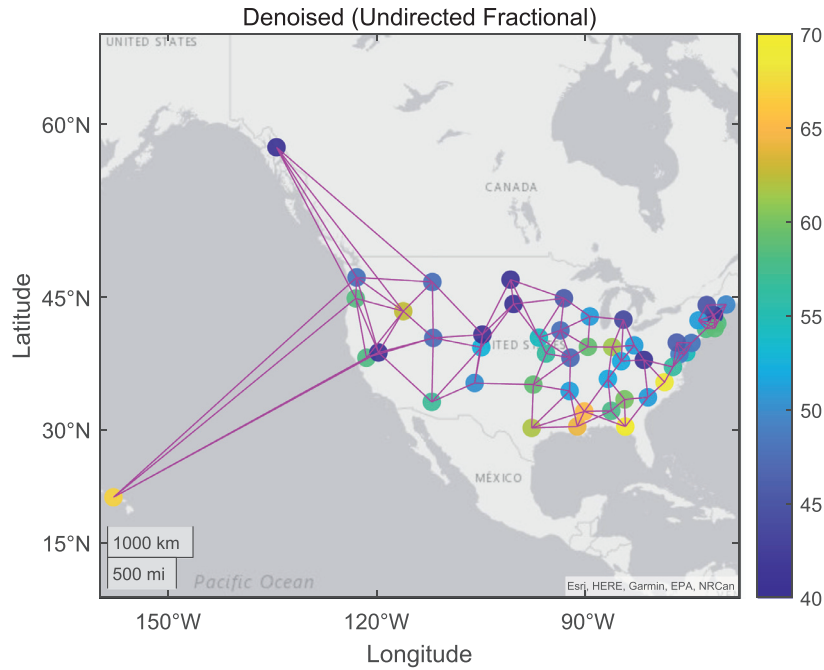
can be calculated as

$$\tilde{f} = U\hat{H}U^*g. \quad (35)$$

where $\hat{H} = \text{diag}\{\hat{h}_d(\lambda_0), \dots, \hat{h}_d(\lambda_{N-1})\}$ and U is the transform matrix coming from classical GFT, SGFRFT, in-degree GFT, Hermitian GFT, and DGRFT. In this experiment, we set $c = 0.02$, $q = 0.5$ and $\alpha = 0.9$. The selection of c was just described above. For parameter q , the imaginary of Γ_q equals to 0 when $q = 0.5$. This value can accelerate the calculation speed in MATLAB. The choice of α is



(c) denoised signal using GFT defined in [4].



(d) denoised signal using SGFRFT [14].

Fig. 5. Continued

through trial and error in most experiments. Note that we can only apply classical GFT [4] and SGFRFT to undirected graphs. The in-degree GFT [24], Hermitian GFT [31] and DGRFT are suitable for directed graphs. To verify the importance of the directionality, we use the above denoising scheme using the same US graph in Fig. 3, where we omit the direction of edges. Figs. 4 and 5 give an example of the noisy, and denoised temperature signals on directed and undirected US graphs respectively. In the directed case, the performance of in-degree GFT, Hermitian GFT and DGRFT is observed in Fig. 4(b)–(d). In the undirected case, the denoising

results after classical GFT and SGFRFT are shown in Fig. 5(c) and (d).

In addition, we calculate the root mean square error (RMSE) between original signal and the denoised signal obtained by using classical GFT, SGFRFT, in-degree GFT, Hermitian GFT, and DGRFT approaches. The whole processing time (time unit = second) is also given. The results are described in Table 2. As the results demonstrate, these five transforms to graph signal denoising perform well, leading to small errors. Among them, all the processing times are relatively short, and the RMSE generated by DGRFT

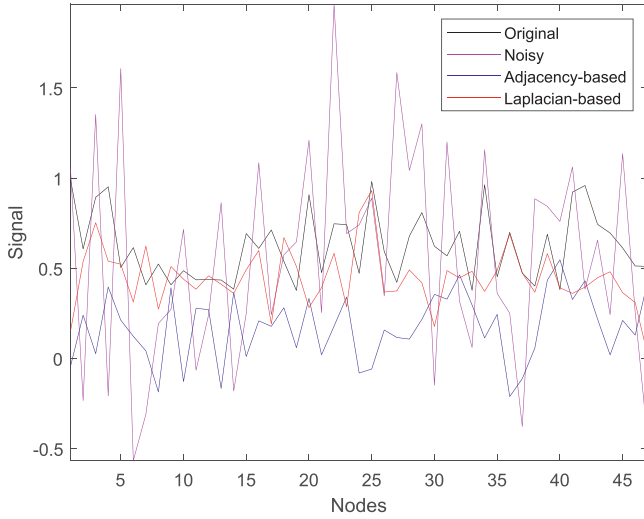


Fig. 6. A sample realization of the original, noisy, and recovered signals of DGFRFT (Laplacian-based) and GFRFT (Adjacency-based).

Table 2
Average RMSE and time cost between denoised signal and original signal.

Graph type	Transform method	RMSE	Processing time (s)
undirected	classical GFT [4]	6.5685	0.000582
undirected	SGFRFT [14]	7.0527	0.001196
directed	in-degree GFT [24]	38.9577	0.001608
directed	Hermitian GFT [31]	6.5657	0.001613
directed	DGFRFT	6.3828	0.004868

is minimal. The RMSE obtained by our method is 6.3828, while this quantity changes to 6.5685, 7.0527, 38.9577, and 6.5657 by using classical GFT and SGFRFT based on an undirected graph, in-degree GFT and Hermitian GFT based on a directed graph respectively. When we compare the results of the three transforms for the directed graph, we can see that the fractional order in DGFRFT produces performance improvements. Furthermore, when both DGFRFT and SGFRFT have fractional order parameter, the existence of edge directivity makes DGFRFT better. In conclusion, the proposed DGFRFT on directed graphs outperforms the other four methods, which highlights its practical usefulness in data denoising.

5.1.2. On large-scale graphs

In this section, we test the effectiveness of DGFRFT on the larger graph. First we consider Wikipedia voting network data used in Leskovec et al. [55,56]. Wikipedia is a free encyclope-

Table 3
Average RMSE and time cost between denoised signal and original signal.

Graph type	Transform method	RMSE	Processing time (s)
directed	Hermitian GFT [31]	47.4008	480.609837
directed	in-degree GFT [24]	4.6719	150.411805
directed	DGFRFT	3.4850	514.359612

dia co-written by volunteers. Some of them were selected as administrators through open discussion or voting. We pulled all of Wikipedia's voting data from its inception through January 2008. The resulting graph contains 7115 vertices and 103,689 edges. Vertices in the graph represent Wikipedia users and a directed edge from vertex i to vertex j represents that user i voted for user j . The data is available in <https://snap.stanford.edu/data/wiki-Vote.html>. There is a natural signal on this graph which is the in-degree of each vertex, i.e. the total number of votes the person receives. We consider signal denoising on this large network to demonstrate the ability of our method. We add Gaussian noise with zero mean and standard deviation $\sigma = 7$. We use the same filter as in Section 5.1.1 with the parameter $c = 1$, $q = 0.5$ and $\alpha = 0.9$. We test three methods on directed graphs: the Hermitian GFT [31], in-degree GFT [24] and our DGFRFT. The results are shown in Table 3. Our method gives the minimal RMSE.

The computational complexity of DGFRFT is $O(N^3)$. In graph fractional domain, there is less discussion about the limit of the scale of the underlying graph structure. This limit also depends on the computer the researchers are using. In [47], the authors show that the Hermitian Laplacian approach can be used on graphs with hundreds of thousands of vertices. The fractional order in DGFRFT gives flexibility in processing data and makes performance improvements possible, mostly without additional computational costs. Thus, we think that DGFRFT can be extended to larger graphs containing hundreds of thousands of vertices.

5.2. Simulation 2: DGFRFT vs. GFRFT

Next, compared with GFRFT [13], which is also applicable to directed graphs, we study a real brain graph to demonstrate the superiority of DGFRFT in denoising tasks. The datasets represent the macaque large-scale visual and sensorimotor area corticocortical connectivity [57]. It has 47 vertices and 505 edges (121 edges are directed). For undirected edge between vertex i and vertex j existing in this dataset, we view it as two directed edges (i, j) and (j, i) , from i to j and j to i respectively. The vertices represent cortical areas and edges represent large corticocortical tracts or functional associations. The data is available in <https://sites.google.com/site/bctnet/>.

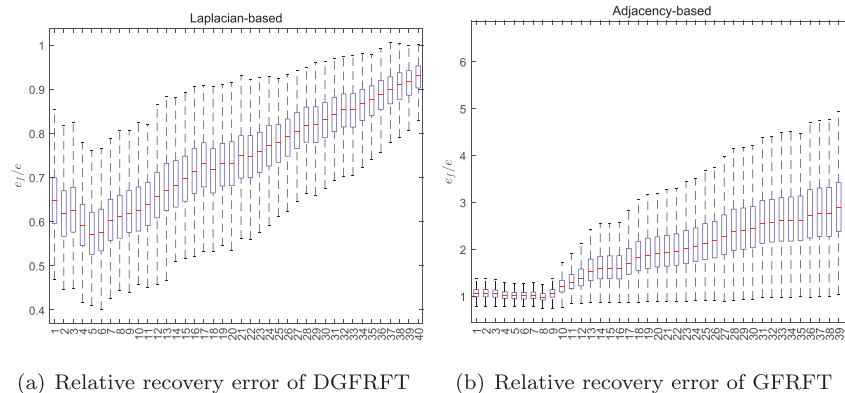


Fig. 7. Relative recovery error of DGFRFT and GFRFT when $\alpha = 0.9$ and $q = 0.2$.

Let U be the orthonormal DGRFT or GFRFT basis. We construct a synthetic graph signal by $x_f = e^{-f}$. We add Gaussian noise to the original signal to obtain noisy signal $g = x_f + n$. We use a series of filter kernels constructed by $H := \text{diag}(h)$, where $h_i = 1[i \leq \ell]$ and ℓ is a tuning parameter to control the number of diagonal elements. The parameter ℓ can be viewed as spectral window size and when $\ell = 47$ the filter kernel is just the identity matrix. The recovered signal is given by

$$\tilde{x}_f = UHU^*g. \quad (36)$$

Fig. 6 shows an example of original, noisy, and recovered signals of DGRFT and GFRFT with window size 5. Moreover, we compute the relative recovery error with respect to the true error. The recovery error is defined as $e_f = \|\tilde{x}_f - x_f\|/\|x_f\|$. The true error is defined as $e = \|n\|/\|x\|$, and the relative recovery error is defined as e_f/e . Fig. 7 are boxplots depicting e_f/e versus ℓ averaged over 1000 Monte-Carlo simulations, and demonstrates the effectiveness of adopting filters along with the proposed two methods. We can see that DGRFT is much more stable than GFRFT. The Jordan decomposition required in GFRFT is numerically unstable, which may be responsible for the large reconstruction errors. Note that the instability problems involved can be solved in some cases by using the multi-precision arithmetic libraries in the computed experiments [58,59]. Alternative decomposition methods, such as Schur decomposition, have also been proposed to solve these instability problems [60]. Therefore, compared with GFRFT, DGRFT has certain advantages when dealing with graph signals whose underlying structure is a directed graph.

6. Conclusions

Signals defined on directed graphs have important practical significance. In this paper, we propose a methodology to conduct graph signal processing on directed graphs in spectral graph fractional domain. First, we introduce a method to construct a new fractional Laplacian matrix for directed graphs and prove that it is a positive semi-definite Hermitian matrix. Equipped with this Hermitian fractional Laplacian matrix, a new transform named DGRFT is presented. Then, to highlight the utility of DGRFT, we propose two basic filtering methods and use them for signal denoising on real data. Finally, the effectiveness of the DGRFT construction is illustrated through experiments on three kinds of directed real-world graphs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Fang-Jia Yan: Conceptualization, Writing – original draft.
Bing-Zhao Li: Writing – review & editing.

Data availability

Data will be made available on request.

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